

Calculus II Lesson 23

1) Find a geometric power series for the function $f(x) = \frac{1}{4 - x}$ centered at 0.

$$\text{Note: } f(x) = \frac{1}{4 - x} = \frac{1/4}{(4 - x)/4} = \frac{1/4}{1 - x/4} = \frac{a}{1 - r}$$

Note: To express $f(x) = \frac{1}{4 - x}$ as a power series, we need to rewrite $f(x) = \frac{1}{4 - x}$ so that it's of the form $\frac{a}{1 - r}$.

a) $a =$ _____ ?

b) $r =$ _____ ?

c) Power Series: $\frac{1}{4 - x} = \sum_{n=0}^{\infty} ar^n =$ _____ ? (Multiple Choice)

A) $\sum_{n=0}^{\infty} 4\left(\frac{x}{4}\right)^n$ B) $\sum_{n=0}^{\infty} \frac{1}{4}\left(\frac{x}{4}\right)^n$ C) $\sum_{n=0}^{\infty} \frac{1}{4}(x)^n$ D) $\sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n$

2) Find a geometric power series for the function $f(x) = \frac{3}{4 - x}$ centered at 0.

Note: $\frac{3}{4 - x} = \frac{(3)/4}{(4 - x)/4} = \frac{3/4}{1 - x/4} = \frac{a}{1 - r}$

Note: To express $f(x) = \frac{3}{4 - x}$ as a power series, we need to rewrite $f(x) = \frac{3}{4 - x}$ so that it's of the form $\frac{a}{1 - r}$.

a) $a =$ _____

b) $r =$ _____

c) Power Series: $\frac{3}{4 - x} = \sum_{n=0}^{\infty} ar^n =$ _____ (Multiple Choice)

A) $\sum_{n=0}^{\infty} 3\left(\frac{x}{4}\right)^n$ B) $\sum_{n=0}^{\infty} \frac{1}{4}\left(\frac{x}{4}\right)^n$ C) $\sum_{n=0}^{\infty} \frac{3}{4}\left(\frac{x}{4}\right)^n$ D) $\sum_{n=0}^{\infty} \left(\frac{4}{4}\right)^n$

3) Find a geometric power series for the function $f(x) = \frac{1}{3-x}$ centered at $c=1$.

To center a function or power series at $x = c$, substitute x with $(x-c)$.

To center a function or power series at $x = 1$, substitute x with $(x-1)$ and make all necessary adjustments.

$$\text{Hence, } f(x) = \frac{1}{3-x} = \frac{1}{3-(x-1+1)} = \frac{1}{3-(x-1)-1} = \frac{1}{2-(x-1)} = \frac{1/2}{2/2-(x-1)/2} = \frac{1/2}{1-\frac{1}{2}(x-1)} = \frac{a}{1-r}$$

Note: To express $f(x) = \frac{1}{3-x}$ as a power series, we need to rewrite $f(x) = \frac{1}{3-x}$ so that it's of the form $\frac{a}{1-r}$.

a) $a =$ _____?

b) $r =$ _____?

c) Power Series: $\frac{1}{3-x} = \sum_{n=0}^{\infty} ar^n =$ _____ (Multiple Choice)

A) $\sum_{n=0}^{\infty} \left(\frac{x-1}{2}\right)^n$ B) $\sum_{n=0}^{\infty} \frac{1}{2}(x-1)^n$ C) $\sum_{n=0}^{\infty} \frac{1}{4}\left(\frac{x-1}{4}\right)^n$ D) $\sum_{n=0}^{\infty} \frac{1}{2}\left(\frac{x-1}{2}\right)^n$

4) Find a geometric power series for the function $f(x) = \frac{1}{1 - 3x}$ centered at $c = 0$.

Note: $\frac{1}{1 - 3x} = \frac{a}{1 - r}$

Note: To express $f(x) = \frac{1}{1 - 3x}$ as a power series, we need to rewrite $f(x) = \frac{1}{1 - 3x}$ so that it's of the form $\frac{a}{1 - r}$.

a) $a =$ _____ ?

b) $r =$ _____ ?

c) Power Series: $\frac{1}{1 - 3x} = \sum_{n=0}^{\infty} ar^n =$ _____ ? (Multiple Choice)

A) $\sum_{n=0}^{\infty} \frac{1}{2}(3x)^n$

B) $\sum_{n=0}^{\infty} (3x)^n$

C) $\sum_{n=0}^{\infty} \frac{1}{4}(3x)^n$

D) $\sum_{n=0}^{\infty} 2(3x)^n$

5) Find a geometric power series for the function $f(x) = \frac{5}{2x - 3}$ centered at $c = -3$.

To center a function or power series at $x = c$, substitute x with $(x - c)$.

To center a function or power series at $x = -3$, substitute x with $(x - (-3)) = (x + 3)$ and make all necessary adjustments.

$$\begin{aligned}\text{Hence, } f(x) &= \frac{5}{2x - 3} = \frac{5}{2(x + 3 - 3) - 3} = \frac{5}{2(x + 3) - 6 - 3} = \frac{5}{2(x + 3) - 9} = \frac{5}{-9 + 2(x + 3)} \\ &= \frac{5/(-9)}{-9/(-9) + 2(x + 3)/(-9)} = \frac{-5/9}{1 - \frac{2}{9}(x + 3)} = \frac{a}{1 - r}\end{aligned}$$

Note: To express $f(x) = \frac{5}{2x - 3}$ as a power series, we need to rewrite $f(x) = \frac{5}{2x - 3}$ so that it's of the form $\frac{a}{1 - r}$.

a) $a =$ _____ ?

b) $r =$ _____ ?

c) Power Series: $\frac{5}{2x - 3} = \sum_{n=0}^{\infty} ar^n =$ _____ ? (Multiple Choice)

A) $\sum_{n=0}^{\infty} \left(\frac{1}{9} \right) \left[\frac{2}{9}(x + 3) \right]^n$

B) $\sum_{n=0}^{\infty} \left(\frac{-1}{9} \right) \left[\frac{2}{9}(x + 3) \right]^n$

C) $\sum_{n=0}^{\infty} \left(\frac{-5}{9} \right) \left[\frac{2}{9}(x + 3) \right]^n$

D) $\sum_{n=0}^{\infty} \left(\frac{-4}{9} \right) \left[\frac{4}{9}(x + 3) \right]^n$

6) Find a geometric power series for the function $f(x) = \frac{3}{3x + 4}$ centered at $c = 0$.

$$\text{Note: } f(x) = \frac{3}{3x + 4} = \frac{3/4}{3x/4 + 4/4} = \frac{3/4}{1 - (-3x/4)} = \frac{a}{1 - r}$$

Note: To express $f(x) = \frac{3}{3x + 4}$ as a power series, we need to rewrite $f(x) = \frac{3}{3x + 4}$ so that it's of the form $\frac{a}{1 - r}$.

a) $a =$ _____

b) $r =$ _____

c) Power Series: $\frac{3}{3x + 4} = \sum_{n=0}^{\infty} ar^n =$ _____

A) $\sum_{n=0}^{\infty} \left(\frac{1}{4}\right) \left[\frac{-3x}{4}\right]^n$

B) $\sum_{n=0}^{\infty} \left(\frac{1}{4}\right) \left[\frac{-3x}{2}\right]^n$

C) $\sum_{n=0}^{\infty} \left(\frac{3}{4}\right) \left[\frac{-3x}{4}\right]^n$

D) $\sum_{n=0}^{\infty} \left(\frac{-3}{4}\right) \left[\frac{3x}{4}\right]^n$

7) Find a power series for the function $f(x) = \frac{x}{x^2 + 2x - 3}$ centered at $c = 0$:

Using Partial Fraction Decomposition: $\frac{x}{x^2 + 2x - 3} = \frac{3/4}{x + 3} + \frac{1/4}{x - 1}$

Hint: Rewrite $\frac{3/4}{x+3}$ as $\frac{a}{1-r}$

Hint: Rewrite $\frac{1/4}{x-1}$ as $\frac{a}{1-r}$

Power Series: _____ (Multiple Choice)

A) $\sum_{n=0}^{\infty} \left[\left(\frac{1}{2} \right) \left(\frac{-x}{3} \right)^n + \left(-\frac{1}{2} \right) x^n \right] = \sum_{n=0}^{\infty} \left(\frac{(-1)^n}{2 \cdot 3^n} - \frac{1}{2} \right) x^n$ B) $\sum_{n=0}^{\infty} \left[\left(\frac{1}{4} \right) \left(\frac{-x}{3} \right)^n + \left(-\frac{1}{4} \right) x^n \right] = \sum_{n=0}^{\infty} \left(\frac{(-1)^n}{4 \cdot 3^n} - \frac{1}{4} \right) x^n$

C) $\sum_{n=0}^{\infty} \left[\left(\frac{1}{5} \right) \left(\frac{-x}{2} \right)^n + \left(-\frac{1}{3} \right) x^n \right] = \sum_{n=0}^{\infty} \left(\frac{(-1)^n}{5 \cdot 2^n} - \frac{1}{3} \right) x^n$ D) $\sum_{n=0}^{\infty} \left[\left(\frac{1}{7} \right) \left(\frac{-x}{4} \right)^n + \left(-\frac{1}{4} \right) x^n \right] = \sum_{n=0}^{\infty} \left(\frac{(-1)^n}{7 \cdot 4^n} - \frac{1}{4} \right) x^n$

8) Find a power series for the function $f(x) = \frac{4}{1-x^2}$ centered at $c = 0$.

Using Partial Fraction Decomposition: $\frac{4}{1-x^2} = \frac{-2}{x-1} + \frac{2}{x+1}$

Hint: Rewrite $\frac{-2}{x-1}$ as $\frac{a}{1-r}$

Hint: Rewrite $\frac{2}{x+1}$ as $\frac{a}{1-r}$

Power Series: _____ (Multiple Choice)

A) $\sum_{n=0}^{\infty} [2x^n + 2(-x)^n] = \sum_{n=0}^{\infty} (2 - 2(-1)^n)x^n$

B) $\sum_{n=0}^{\infty} [3x^n + 2(-x)^n] = \sum_{n=0}^{\infty} (3 - 2(-1)^n)x^n$

C) $\sum_{n=0}^{\infty} [4x^n + 3(-x)^n] = \sum_{n=0}^{\infty} (4 - 3(-1)^n)x^n$

D) $\sum_{n=0}^{\infty} [x^n + 3(-x)^n] = \sum_{n=0}^{\infty} (1 - 3(-1)^n)x^n$

